Empirical Evaluation of Transfer Learning Stages on CIFAR-10 with Limited Training Data

Ammar Shaikh and Khalid Shaikh

Abstract

Deep learning models achieve high performance in artificial intelligence applications such as computer vision, natural language processing, and speech recognition. However, training deep neural networks from scratch requires large labeled datasets and high computational resources, which are not always available in real-world scenarios. Transfer learning is widely used to overcome this limitation by reusing knowledge from pretrained neural networks. Although many studies discuss transfer learning, fewer works experimentally evaluate how different stages of the transfer learning pipeline contribute to performance improvement when training data is limited.

This paper presents an experimental evaluation of a structured transfer learning framework consisting of dataset preparation, pretrained model selection, feature extraction, and adaptive fine-tuning. Experiments were conducted on the CIFAR-10 dataset using a limited number of training samples to simulate a data-constrained environment. Three models were compared: a convolutional neural network trained from scratch, transfer learning with feature extraction, and transfer learning with fine-tuning.

Results show that transfer learning significantly improves accuracy compared to training from scratch, and fine-tuning further enhances performance. Additional analysis using training curves, confusion matrices, and per-class accuracy confirms the effectiveness of the proposed framework. The experimental results validate that applying transfer learning in a structured pipeline improves deep learning performance in data-limited conditions.

Keywords

Deep Learning, Transfer Learning, Fine-Tuning, Feature Extraction, CIFAR-10, ResNet, Experimental Evaluation

I. Introduction

Deep learning has become one of the most important technologies in modern artificial intelligence. Over the past decade, deep neural networks have achieved remarkable performance in applications such as image classification, object detection, speech recognition, and natural language processing. Architectures such as convolutional neural networks (CNNs) and transformer-based models are capable of learning complex patterns from large datasets and have significantly improved the accuracy of machine learning systems.

Despite these advancements, training deep neural networks from scratch requires very large labeled datasets and high computational resources. Modern deep learning models often contain millions of parameters that must be optimized during training. As a result, successful training usually requires powerful GPUs and extensive annotated data. In many real-world applications, such as medical image analysis or industrial inspection, collecting large labeled datasets is expensive and time consuming.

Transfer learning has emerged as an effective technique to address this limitation. Instead of training a model from scratch, transfer learning uses pretrained neural networks that have already learned useful feature representations from large datasets such as ImageNet. These learned features can then be reused and adapted for new tasks that have limited data. Transfer learning reduces training time, improves convergence, and often produces higher accuracy when the available dataset is small.

Although transfer learning is widely used, many studies describe the method conceptually without experimentally analyzing how each stage of the transfer learning pipeline contributes to performance improvement. In particular, there is limited experimental comparison between training from scratch, feature extraction, and fine-tuning under data-limited conditions. Understanding the contribution of each stage is important for designing efficient deep learning systems.

In this work, a structured transfer learning framework is evaluated experimentally. The framework consists of dataset preparation, pretrained model selection, feature extraction, and adaptive fine-tuning. Experiments are performed on the CIFAR-10 dataset using a limited number of training samples to simulate real-world data constraints. The results demonstrate how each stage of the framework improves model performance and confirms the effectiveness of transfer learning in data-limited environments.

II. Research Gap

Transfer learning has been widely studied in deep learning research and has been successfully applied in many applications such as image classification, object detection, and natural language processing. Previous studies have shown that pretrained neural networks can improve performance when the available training data is limited. Models such as AlexNet, ResNet, VGG, and EfficientNet trained on large datasets like ImageNet are commonly used as feature extractors for new tasks.

However, many existing studies focus on applying transfer learning directly without experimentally analyzing how different stages of the transfer learning pipeline contribute to performance improvement. In many cases, transfer learning is used as a single step, and the individual effects of feature extraction and fine-tuning are not evaluated separately. As a result, it is not always clear how much performance improvement comes from each stage of the transfer learning process.

Another limitation in existing research is that many experiments are performed using large datasets, where training from scratch may still produce acceptable results. In real-world scenarios, especially in medical imaging, industrial inspection, and small-scale applications, the available training data is often very limited. Under such conditions, the behavior of transfer learning methods can be different, and the benefit of each stage of the transfer learning pipeline needs to be analyzed carefully.

In addition, several studies report overall accuracy improvement using transfer learning, but fewer works provide detailed analysis using multiple evaluation methods such as training curves, confusion matrices, and per-class accuracy. These additional analyses are important to understand how transfer learning affects model convergence, generalization, and classification errors.

Therefore, there is a need for an experimental study that evaluates transfer learning in a structured way by separating the main stages of the pipeline, including training from scratch, feature extraction, and fine-tuning, and comparing their performance in a data-limited environment.

In this paper, a structured transfer learning framework is evaluated experimentally using the CIFAR-10 dataset with a limited number of training samples. The study compares three configurations: training a convolutional neural network from scratch, transfer learning using feature extraction, and transfer learning with fine-tuning. The goal is to analyze how each stage of the framework contributes to performance improvement and to validate the effectiveness of the transfer learning pipeline in data-limited conditions.

III. Literature Review

Transfer learning has become an essential technique in deep learning, particularly when training data is limited. Instead of training neural networks from scratch, pretrained models can be reused and adapted to new tasks. Several studies have shown that transfer learning improves performance, reduces training time, and enables deep learning models to generalize better across different datasets.

Pan and Yang [1] presented one of the earliest comprehensive surveys on transfer learning and defined it as the process of improving learning performance in a target domain by transferring knowledge from a related source domain. Their work categorized transfer learning into different types, including feature representation transfer and parameter transfer.

Krizhevsky et al. [2] introduced AlexNet, a deep convolutional neural network that achieved breakthrough performance on the ImageNet dataset. This work demonstrated that deep neural networks trained on large datasets can learn powerful feature representations that can later be reused for other tasks.

Deng et al. [3] developed the ImageNet dataset, which contains millions of labeled images across thousands of categories. ImageNet has played a major role in the development of transfer learning, as many deep neural networks are pretrained on this dataset before being applied to smaller datasets.

Simonyan and Zisserman [4] proposed the VGG network architecture, which showed that increasing network depth can significantly improve

classification accuracy. VGG models are widely used in transfer learning due to their ability to learn general visual features.

He et al. [5] introduced the ResNet architecture, which uses residual connections to enable very deep neural networks to be trained effectively. ResNet models are commonly used for transfer learning because they provide strong feature representations and high accuracy.

Yosinski et al. [6] studied the transferability of features learned by deep neural networks and showed that early layers learn general features such as edges and textures, while later layers learn task-specific features. Their results support the use of feature extraction and fine-tuning in transfer learning.

Howard et al. [7] proposed MobileNet, a lightweight neural network architecture designed for mobile and embedded devices. MobileNet models are often used in transfer learning when computational resources are limited.

Tan and Le [8] introduced EfficientNet, which improves performance by scaling network depth, width, and resolution in a balanced manner. EfficientNet models have been successfully used for transfer learning due to their efficiency and high accuracy.

Kornblith et al. [9] evaluated the transferability of deep neural network features and showed that models with higher performance on ImageNet tend to transfer better to new datasets. Their work highlights the importance of pretrained models in transfer learning.

Donahue et al. [10] demonstrated that features extracted from pretrained convolutional neural networks can be used as generic representations for different visual recognition tasks. Their study showed that transfer learning can significantly improve performance even with small datasets.

Long et al. [11] proposed Deep Adaptation Networks, which reduce the difference between source and target domains to improve transfer learning performance. Their work shows that domain adaptation plays an important role when datasets are different.

Weiss et al. [12] provided a survey of transfer learning methods and discussed their applications in various machine learning tasks. Their study emphasized that transfer learning is especially useful when labeled data is limited.

Zhuang et al. [13] presented a comprehensive survey on transfer learning and discussed challenges such as domain mismatch, negative transfer, and dataset similarity. These challenges affect the performance of transfer learning methods.

Goodfellow et al. [14] introduced Generative Adversarial Networks, which improved representation learning in deep neural networks. Better feature representations can improve transfer learning performance.

LeCun, Bengio, and Hinton [15] discussed the foundations of deep learning and showed that hierarchical feature learning allows neural networks to generalize across different tasks.

Sharif Razavian et al. [16] showed that features extracted from pretrained CNN models can serve as strong baseline representations for many computer vision tasks without requiring extensive training.

Girshick et al. [17] proposed R-CNN for object detection, demonstrating that pretrained convolutional networks can be reused for different tasks with improved accuracy.

Ren et al. [18] introduced Faster R-CNN, which further improved detection performance by integrating deep learning and region proposal networks.

Dosovitskiy et al. [19] proposed Vision Transformer, which applies transformer architecture to image classification and shows that pretrained models can achieve strong performance when transferred to new tasks.

Radford et al. [20] introduced CLIP, a multimodal model trained on large image-text datasets, showing that pretrained models can generalize to many different tasks without task-specific training.

Although many studies demonstrate the effectiveness of transfer learning, fewer works experimentally compare training from scratch, feature extraction, and fine-tuning under the same limited-data conditions. In addition, detailed evaluation using training curves, confusion matrices, and per-class accuracy is not always provided.

IV. Proposed Framework

Based on the analysis of existing transfer learning techniques, this study evaluates a structured transfer

learning framework for improving deep learning performance in data-limited environments. The framework is derived from commonly used transfer learning workflows and is experimentally validated to analyze how each stage contributes to model accuracy.

The proposed framework consists of four main stages: dataset preparation, pretrained model selection, feature extraction, and adaptive fine-tuning. These stages are applied sequentially and evaluated experimentally to measure their effect on model performance.

The goal of this framework is not to introduce a new algorithm, but to systematically analyze how transfer learning improves performance when the available training data is limited.

A. Dataset Preparation

The first stage of the framework involves preparing the dataset for training. In real-world applications, large labeled datasets are not always available, which makes training deep neural networks difficult.

To simulate a data-limited environment, the CIFAR-10 dataset was used with a reduced number of training samples. CIFAR-10 contains 60,000 images divided into 10 classes, but only 1000 training images were selected randomly for training. The full test dataset was used for evaluation.

All images were resized to 224 × 224 pixels to match the input size required by pretrained convolutional neural network models. Normalization was applied to improve training stability.

This stage represents the dataset preparation step of the proposed framework.

B. Pretrained Model Selection

The second stage of the framework involves selecting a suitable pretrained deep learning model. Pretrained models trained on large datasets such as ImageNet contain useful feature representations that can be reused for new tasks.

In this study, the ResNet18 architecture pretrained on the ImageNet dataset was used. ResNet models are widely used for transfer learning because residual connections allow deep networks to learn strong feature representations without degradation.

The pretrained model provides a strong starting point for training when the available dataset is small.

C. Feature Extraction

Feature extraction is the third stage of the framework. In this step, the early layers of the pretrained network are frozen so that their parameters are not updated during training.

These early layers contain general features such as edges, textures, and shapes that are useful for many image classification tasks. By freezing these layers, the model can reuse previously learned knowledge while only training the final classification layer.

In this study, the convolutional layers of the pretrained ResNet18 model were frozen, and only the final fully connected layer was trained using the target dataset.

This stage allows the model to learn task-specific classification without changing the general feature representations.

D. Adaptive Fine-Tuning

The final stage of the framework is adaptive fine-tuning. In this step, some of the deeper layers of the pretrained model are unfrozen and retrained together with the classification layer.

Fine-tuning allows the model to adjust previously learned features to better match the target dataset. This can improve performance when the target dataset differs from the dataset used during pretraining.

In this study, the deeper layers of the ResNet18 model were unfrozen while the earlier layers remained frozen. The model was then trained using the limited training dataset.

This stage represents the complete transfer learning pipeline and is expected to produce the highest accuracy.

V. Experimental Evaluation

A. Dataset

To validate the effectiveness of the proposed transfer learning framework, experiments were conducted using the CIFAR-10 dataset. CIFAR-10 is a standard benchmark dataset in computer vision

consisting of 60,000 color images divided into 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

In order to simulate a data-limited environment, only 1000 training samples were randomly selected from the original training dataset. The full test dataset containing 10,000 images was used for evaluation.

All input images were resized to 224 × 224 pixels to match the input size required by the pretrained ResNet18 model. Normalization was applied to improve training stability.

This setup represents real-world scenarios where large labeled datasets are not available.

B. Experimental Setup

Three different models were evaluated to analyze the effect of each stage of the proposed framework.

1) CNN from Scratch

A simple convolutional neural network was trained from scratch using the limited training dataset. This model serves as the baseline and represents training without transfer learning.

2) Transfer Learning with Feature Extraction

A pretrained ResNet18 model trained on ImageNet was used.

All convolutional layers were frozen, and only the final fully connected layer was trained.

This configuration corresponds to the feature extraction stage of the proposed framework.

3) Transfer Learning with Fine-Tuning

The pretrained ResNet18 model was used again, but deeper layers were unfrozen and retrained along with the final classification layer.

This configuration represents the adaptive fine-tuning stage of the proposed framework.

All models were trained using the Adam optimizer with a learning rate of 0.001.

Cross-entropy loss was used as the loss function. Each model was trained for 10 epochs with a batch size of 32 using GPU acceleration in Google Colab.

C. Evaluation Metrics

Model performance was evaluated using classification accuracy on the test dataset.

In addition to overall accuracy, the following metrics were analyzed:

- Training and validation accuracy curves
- Training and validation loss curves
- Confusion matrix
- Per-class accuracy

These metrics provide detailed insight into model performance, convergence behavior, and classification errors.

Accuracy is calculated as:

Accuracy = Correct Predictions / Total Samples × 100

D. Results

Table 1 shows the final accuracy obtained for each model.

Table 1. Model Performance Comparison

Model	Test Accuracy (%)
CNN from Scratch	30.56
Transfer Learning (Feature Extraction)	72.66
Transfer Learning + Fine-Tuning	74.91

The results show that the model trained from scratch achieved very low accuracy due to the limited dataset size. Using transfer learning significantly improved performance because the pretrained model already contains useful feature representations learned from a large dataset. Fine-tuning further improved accuracy by allowing deeper layers to adapt to the target dataset. These results confirm that each stage of the proposed framework contributes to performance improvement.

Figure 3 shows the comparison of final test accuracy for all models. Transfer learning significantly

improves performance compared to training from scratch, and fine-tuning provides the best accuracy.

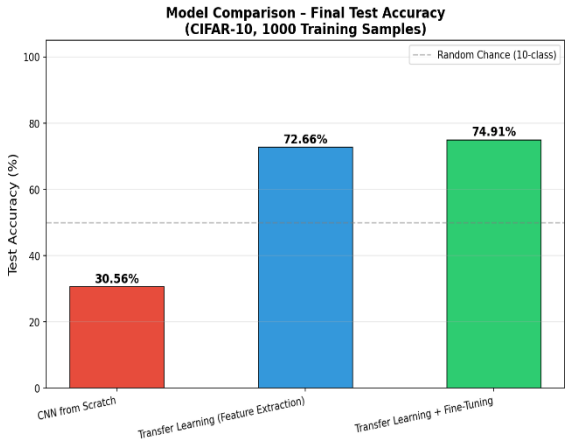

Figure 3. Accuracy comparison of different models.

E. Training Curve Analysis

The training and validation accuracy curves show that the model trained from scratch struggles to learn with limited data. Transfer learning improves convergence speed and produces higher accuracy. Fine-tuning provides the best performance because the model can adapt both general and task-specific features. The loss curves also show that transfer learning reduces training loss faster than training from scratch.

Figure 1 shows the training and validation accuracy curves for all three models. The model trained from scratch converges slowly and achieves low accuracy, while transfer learning significantly improves convergence speed. Fine-tuning achieves the highest accuracy.

Figure 2 shows the training and validation loss curves. The model trained from scratch has higher loss compared to transfer learning models. Fine-tuning results in the lowest loss, indicating better optimization.

Figure 1. Training and validation accuracy curves for different models.

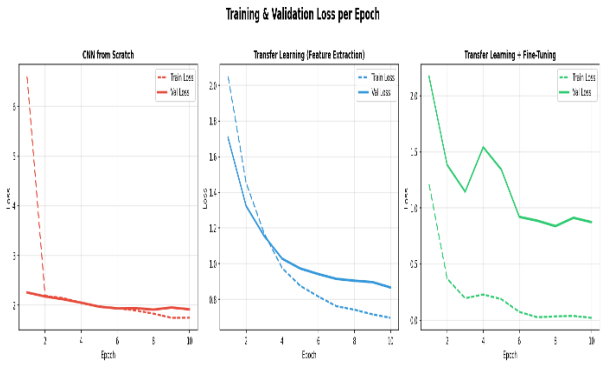

Figure 2. Training and validation loss curves.

F. Confusion Matrix Analysis

Confusion matrix results show that the model trained from scratch produces many classification errors across multiple classes.

Transfer learning reduces the number of incorrect predictions, and fine-tuning further improves classification accuracy.

Classes such as ship, truck, and automobile show significant improvement when the full transfer learning framework is applied.

Figure 4 shows the confusion matrices for the three models. The model trained from scratch produces more classification errors, while transfer learning reduces errors across most classes. Fine-tuning further improves classification accuracy.

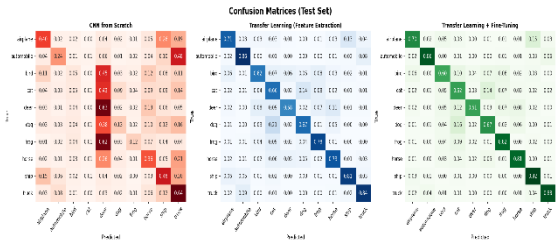

Figure 4. Confusion matrices for all models.

G. Per-Class Accuracy Analysis

Per-class accuracy results show that the model trained from scratch performs poorly for several classes due to insufficient training data.

Feature extraction improves accuracy for most classes, while fine-tuning provides the best overall performance.

These results demonstrate that the proposed framework improves generalization across different classes.

Figure 5 shows per-class accuracy for the fine-tuned model. The results indicate that transfer learning improves performance across all classes, especially for complex categories such as ship, truck, and automobile.

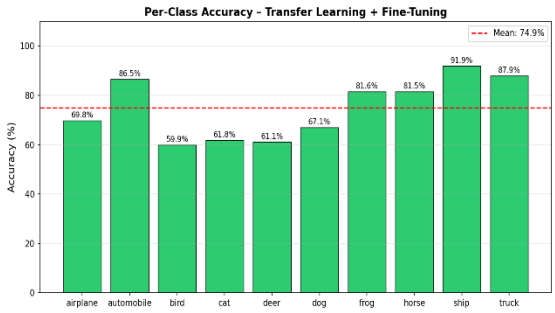

Figure 5. Per-class accuracy for fine-tuned model.

H. Validation of Proposed Framework

The experiment validates the proposed transfer learning framework by showing that Training from scratch performs poorly in data-limited conditions Feature extraction significantly improves performance. Fine-tuning further increases accuracy

Therefore, the experimental results confirm that applying the transfer learning pipeline step-by-step leads to better model performance.

VI. Discussion

The purpose of this study was to experimentally evaluate a structured transfer learning framework in a data-limited environment. Previous studies have shown that transfer learning improves deep learning performance, but the contribution of each stage of the transfer learning pipeline is not always analyzed separately. In this work, training from scratch, feature extraction, and fine-tuning were evaluated under the same conditions to understand how each stage affects model accuracy.

The experimental results clearly show that training a convolutional neural network from scratch performs poorly when the dataset is small. The baseline model achieved only 30.56% accuracy, which indicates that deep neural networks require large amounts of data to learn useful features. This result confirms the limitation of training deep models without pretrained knowledge.

When transfer learning with feature extraction was applied, accuracy increased significantly to 72.66%. This improvement occurs because the pretrained ResNet18 model already contains general feature representations learned from the ImageNet dataset. Freezing the early layers allows the model to reuse these features while training only the final classification layer.

Fine-tuning further improved the accuracy to 74.91%. By unfreezing deeper layers, the model was able to adapt its learned features to the target dataset. Although the improvement compared to feature extraction is smaller, the results show that fine-tuning helps the model achieve better generalization.

Training and validation curves also support these observations. The model trained from scratch shows slow convergence and unstable accuracy, while transfer learning models converge faster and achieve higher performance. Loss curves indicate that transfer learning reduces training loss more effectively than training from scratch.

Confusion matrix analysis shows that the baseline model produces many classification errors across different classes. Transfer learning reduces these errors, and fine-tuning further improves classification accuracy for difficult classes such as ship, truck, and automobile. Per-class accuracy results also confirm that the full transfer learning pipeline provides more consistent performance across all categories.

These results validate the proposed framework by demonstrating that each stage of the transfer learning pipeline contributes to performance improvement. Dataset preparation enables training under limited data conditions, pretrained model selection provides useful feature representations, feature extraction improves accuracy without large training cost, and fine-tuning further adapts the model to the target dataset.

Therefore, the experimental results support the effectiveness of the proposed framework for improving deep learning performance in data-limited environments.

VII. Future Work

Although the proposed transfer learning framework shows significant improvement in performance under data-limited conditions, several extensions can be explored in future work.

In this study, experiments were conducted using the CIFAR-10 dataset with a reduced number of training samples. Future work can evaluate the framework on other datasets such as CIFAR-100, ImageNet subsets, or domain-specific datasets like medical images to analyze the behavior of transfer learning in different scenarios.

Only one pretrained architecture, ResNet18, was used in this study. Future research can compare different pretrained models such as VGG, MobileNet, EfficientNet, and Vision Transformer to understand how model architecture affects transfer learning performance.

Another possible extension is to study the effect of different fine-tuning strategies. In this work, only deeper layers were unfrozen, but future studies can evaluate partial fine-tuning, full fine-tuning, and layer-wise learning rate adjustment.

Data augmentation techniques can also be applied to improve performance in data-limited environments. Combining transfer learning with augmentation may further increase accuracy and reduce overfitting.

Finally, more evaluation metrics such as precision, recall, F1-score, and training time can be analyzed to provide a more detailed comparison of transfer learning methods.

VIII. Conclusion

In this paper, a structured transfer learning framework was experimentally evaluated to analyze its effectiveness in improving deep learning performance under data-limited conditions. The framework consists of dataset preparation, pretrained model selection, feature extraction, and adaptive fine-tuning.

Experiments were conducted on the CIFAR-10 dataset using only 1000 training samples to simulate a limited-data environment. Three configurations were compared: training a convolutional neural network from scratch, transfer learning using feature extraction, and transfer learning with fine-tuning.

The results showed that training from scratch achieved very low accuracy, while transfer learning significantly improved performance. Feature extraction increased accuracy from 30.56% to 72.66%, and fine-tuning further improved accuracy to 74.91%. Training curves, loss curves, confusion matrices, and per-class accuracy analysis confirmed that the full transfer learning pipeline provides the best performance.

These results demonstrate that each stage of the proposed framework contributes to performance improvement. The study confirms that transfer learning is highly effective when training data is limited, and that applying transfer learning in a structured pipeline leads to better accuracy and faster convergence.

Therefore, the experimental evaluation validates the proposed framework and shows that transfer learning is a practical and efficient approach for improving deep learning performance in real-world data-constrained environments.

IX. References

[1] S. J. Pan and Q. Yang, "A Survey on Transfer Learning," IEEE Transactions on Knowledge and Data Engineering, vol. 22, no. 10, pp. 1345–1359, 2010.

[2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," Advances in Neural Information Processing Systems (NIPS), 2012.

[3] J. Deng, W. Dong, R. Socher, L. Li, K. Li, and L. Fei-Fei, "ImageNet: A Large-Scale Hierarchical Image Database," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2009.

[4] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," International Conference on Learning Representations (ICLR), 2015.

[5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[6] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How Transferable Are Features in Deep Neural Networks?" Advances in Neural Information Processing Systems (NIPS), 2014.

[7] A. G. Howard et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv preprint arXiv:1704.04861, 2017.

[8] M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for

Convolutional Neural Networks," International Conference on Machine Learning (ICML), 2019.

[9] S. Kornblith, J. Shlens, and Q. V. Le, "Do Better ImageNet Models Transfer Better?" IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2019.

[10] J. Donahue et al., "DeCAF: A Deep Convolutional Activation Feature for Generic Visual Recognition," International Conference on Machine Learning (ICML), 2014.

[11] M. Long, Y. Cao, J. Wang, and M. Jordan, "Learning Transferable Features with Deep Adaptation Networks," International Conference on Machine Learning (ICML), 2015.

[12] K. Weiss, T. M. Khoshgoftaar, and D. Wang, "A Survey of Transfer Learning," Journal of Big Data, 2016.

[13] F. Zhuang et al., "A Comprehensive Survey on Transfer Learning," Proceedings of the IEEE, vol. 109, no. 1, pp. 43–76, 2021.

[14] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning, MIT Press, 2016.

[15] Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," Nature, vol. 521, pp. 436–444, 2015.

[16] A. Sharif Razavian, H. Azizpour, J. Sullivan, and S. Carlsson, "CNN Features Off-the-Shelf: An Astounding Baseline for Recognition," IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2014.

[17] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2014.

[18] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," Advances in Neural Information Processing Systems, 2015.

[19] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," International Conference on Learning Representations (ICLR), 2021.

[20] A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," International Conference on Machine Learning (ICML), 2021.